

Uncertainty**6.1 Gambles and Contingent Commodities**

The outcome of an uncertain situation is referred to as a state of the world. Contingent commodities are commodities whose level depends on which state of the world occurs.

Budget Constraint

The budget constraint for contingent commodities is graphed on an axis with consumption in one state of the world on the horizontal axis, and consumption in the state of the world is not that event on the vertical axis (the two conditions must be complete). The endowment point of the budget constraint is the point where no bet is placed. In general, the slope of the budget constraint is $-N/H$, where H is the change in consumption if the contingency on the horizontal axis occurs, and N is the change in consumption if the contingency on the vertical axis occurs.

Probability and Expected Value

The probability of a given state of the world is a measure of the likelihood that it occurs.

The expected value of X is the value of X that occurs "on average." The expected value is found by weighting the value of X in each state of the world by the probability of that state of the world occurring. $E[X] = p \cdot X_1 + (1-p) \cdot X_2$

The Fair Odds Line

A gamble is actuarially fair if the expected monetary gain equals zero. The budget constraint that reflects the opportunities presented by an actuarially fair gamble is called the fair odds line. The slope of the fair odds line is minus the probability of the contingency on the horizontal axis divided by the probability of the contingency on the vertical axis. The expected value of consumption is equal at every bundle on a fair odds line.

The odds of two events occurring is the ratio of their probabilities. If the probability of rain today is $1/5$ and the probability of no rain is $4/5$, then the odds of rain are 1 to 4. If the probability of one event is p and the probability of a second event is $(1-p)$, the odds of the first event are $p/(1-p)$.

Preferences

Indifference lines are drawn based on preferences in answer to the question, "which bundle of contingent commodities would you prefer?" The shape of the indifference curves depend on the individuals attitude towards risk.

1. Risk Averse: A risk averse person will not bet when offered an actuarially fair gamble. A 45 degree line traces out the locus of all possible certain consumption levels. This is referred to as the certainty line. For a risk-averse individual, the slope of each indifference curve as it intersects the certainty line is minus the odds in favor of the event on the horizontal axis. (i.e. It is tangent to the fair odds line at the 45 degree line). The



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2. Risk Loving: A risk loving person is an individual who prefers an uncertain prospect with a particular expected value to a certainty with the same expected value. The curves are also tangent to the fair odds line whenever the certainty and a fair odds line intersect, but they are bowed outward to the origin.

3. Risk Neutral: A person is said to be risk neutral if he is indifferent among all alternatives with the same expected value. The indifference curve for a risk-neutral person coincides with the fair odds line. The indifference map consists of parallel lines. The slope of each is minus the odds.

Equilibrium

As always, the equilibrium is the intersection of the budget constraint and indifference curve. When confronted with an actuarially fair bet, a risk-averse individual maximizes utility at the endowment point – i.e. the bet is not accepted. A risk-loving individual has indifference curves that



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6.2 Some Applications of Contingent Commodities

Risk Premia

In general, more people are risk averse than risk loving. An important piece of evidence for this effect is that riskier assets tend to pay higher rates of return than safe assets. (e.g. A government bond vs. a risky company bond.)

A risk premium is the extra return on an investment to compensate for risk. The risk premium is the distance between the utility curve that goes through the risky bet, and the utility curve representing the safe level of utility. [pg. 174]

The Role of Diversification

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Tax Evasion

Tax evasion is the failure to pay taxes that are legally due. The decision to take the risk to evade taxes can be analyzed using these uncertainty methods. The decision to evade taxes depends on the tax rate, the penalty rate, and the likelihood of getting audited (p).

Along the 45 degree ray from the origin the MRS of each indifference curve is $p(1-p)$, where p is the probability of an audit. On the other hand, the slope of the budget constraint is minus the marginal tax rate (t) divided by the penalty rate (f). The equilibrium is where the indifference curve hits the budget constraint line – this equilibrium tells whether the individual will pay full taxes or not.

Designing Policy toward Evasion

A risk averse person will choose not to accept an actuarially fair gamble. If the penalty rate (f) and the probability of an audit (p) are set so that the expected gain from getting away with underreporting a dollar of income equals the expected loss from getting caught, then a risk-averse person will not underreport any income. Mathematically, this means:

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6.3 Insurance

A risk-averse person would not accept an actuarially fair gamble, but sometimes they don't have a choice – some risks just inherently exist. Risk averse people will pay to get rid of the risks they have to face.

Fair Insurance

The premium of the insurance policy is the price of obtaining insurance coverage. Actuarially fair insurance is when the premium equals the expected payout by the insurance provider. (The probability of the event happening times the cost if it does happen.) For an actuarially fair insurance policy, the premium for \$1 worth of insurance is simply the probability of the "bad" state of the world occurring. (e.g. if the probability of an injury is 1/5 then a person can buy \$1 of actuarially fair insurance for \$0.20.)

Budget Constraint with Fair Insurance

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Preferences

The shape of the indifference curves depend on a person's tastes. If a person is risk averse, the indifference curves exhibit a diminishing marginal rate of substitution (bowed toward the origin), and on points along a 45 degree from the origin, the marginal rate of substitution is $p/(1-p)$.

Equilibrium Amount of Fair Insurance

When insurance is actuarially fair, the equilibrium occurs where the consumption in the case of the "bad" event is equal to the consumption in the case of "good" event. The individual is fully insured in the sense that consumption is the same in every state of the world.

Just as a capital market allows an individual to spread consumption over different periods of time, an insurance market allows an individual to spread consumption over different states of the world.

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Changing the Premium

When the insurance policy is actuarially unfair, even a risk-averse individual purchases less than full insurance. Intuitively, when the premium is higher than the expected payoff, it is rational for an individual to assume some of the risk in return for reducing payments to the insurance provider.

Changing the Probability of a Lawsuit

The probability of an accident occurring does not affect the budget constraint. However, it does affect the indifference curves. When the probability of a lawsuit (or an accident) increases, tastes for the contingent commodities change. The indifference curves become steeper.

The Importance of Insurance

There are many examples of insurance that are not purchased through formal contracts. For example, a person might buy a safer car even though it's more expensive as insurance against an accident.

6.4 Decision Making with Many Uncertain Outcomes: von Neumann-Morgenstern Utility Decision Trees

A decision tree is a schematic representation of a choice problem that shows the possible outcome and how they are related to current actions. This